

HOME LAB NETWORK MONITORING USING VMware Workstation, WINDOWS SERVER 2022, KALI LINUX, AND WIRESHARK

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1. Objective

The objective of this project is to create a home lab environment using VMware Workstation with Windows Server 2022 and Kali Linux virtual machines. Wireshark is used to capture and analyze network traffic, including ICMP, DNS, and HTTP communications. Additional protocol analysis is performed to understand network behavior and packet structures.

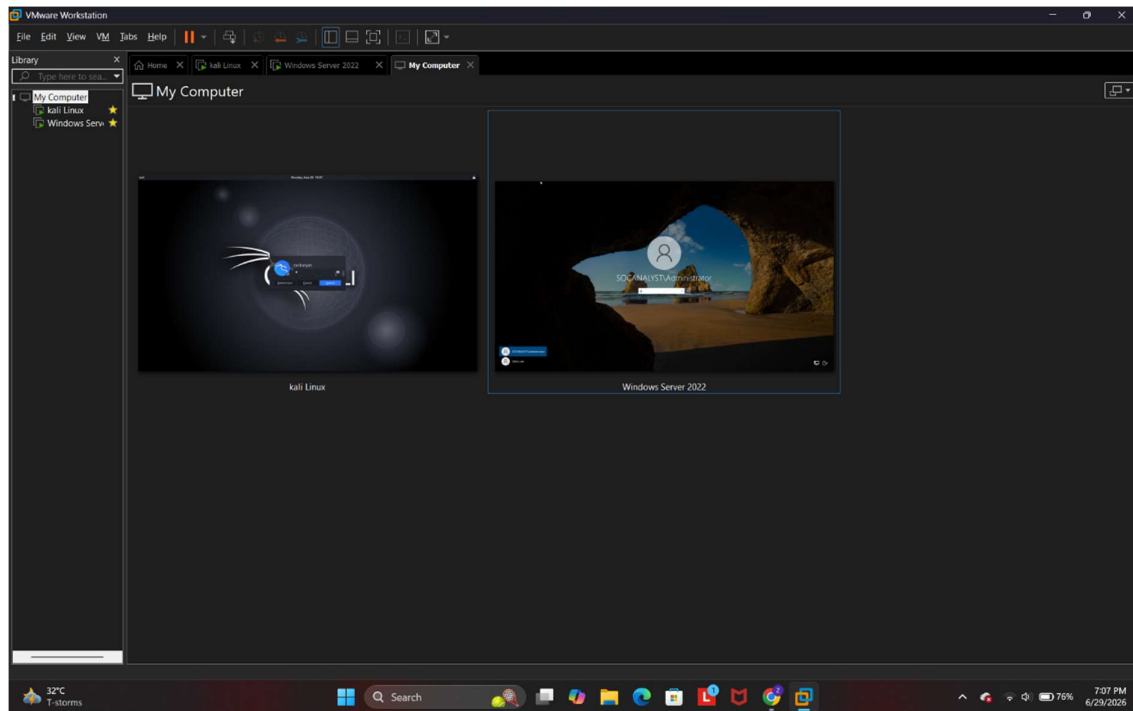
2. Lab Environment

Software Used

1. VMware Workstation
2. Windows Server 2022
3. Kali Linux
4. Wireshark
5. Nmap

Virtual Machine Setup

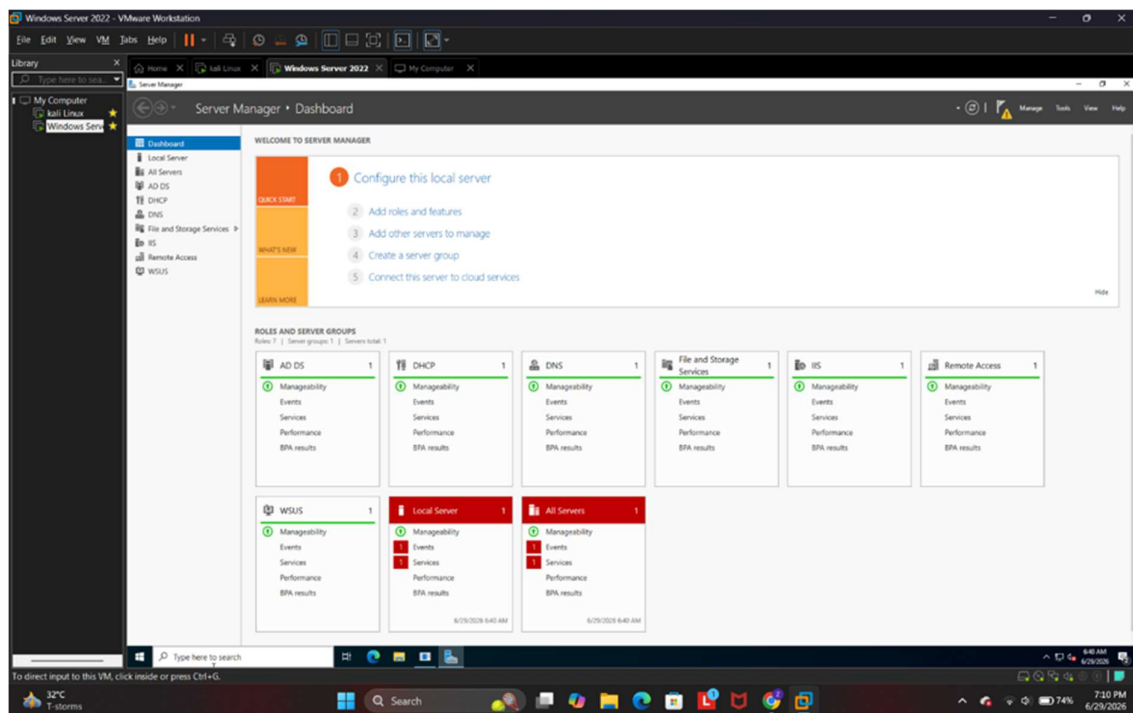
The lab environment consists of two virtual machines connected through a virtual network.



Description:

The VirtualBox environment showing Windows Server 2022 and Kali Linux virtual machines successfully configured and running.

3. Windows Server 2022 Installation



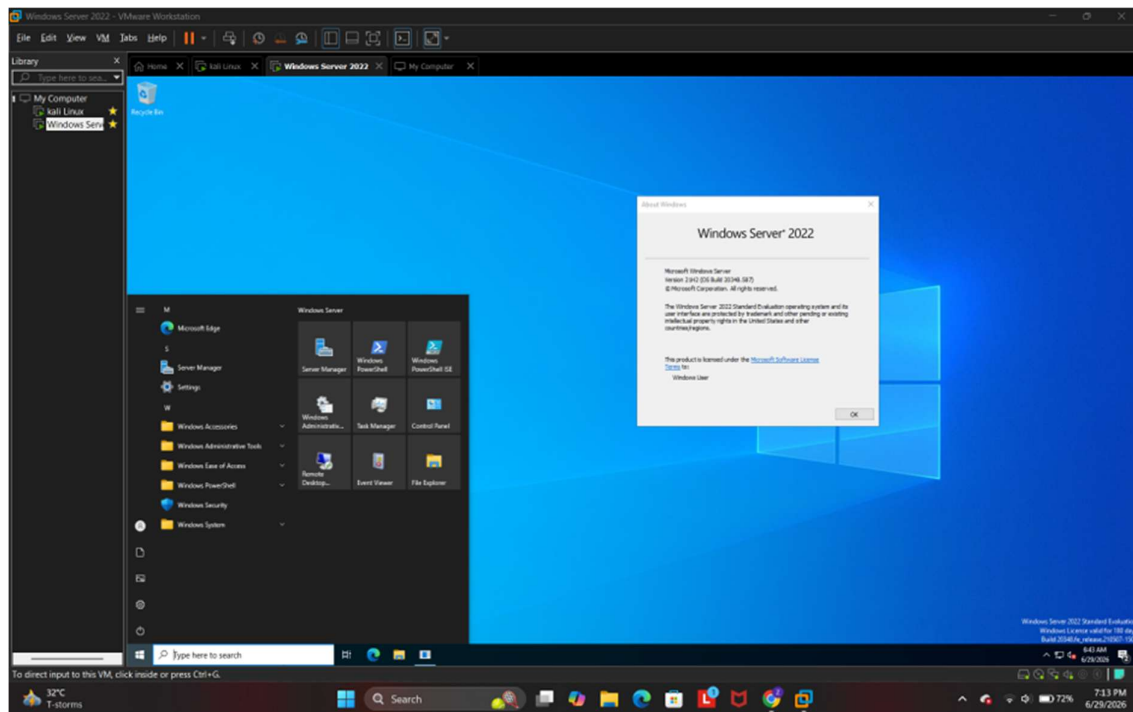
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Figure 2: Windows Server Desktop

Description:

Windows Server 2022 installed successfully.

Figure 3: Windows Version Verification

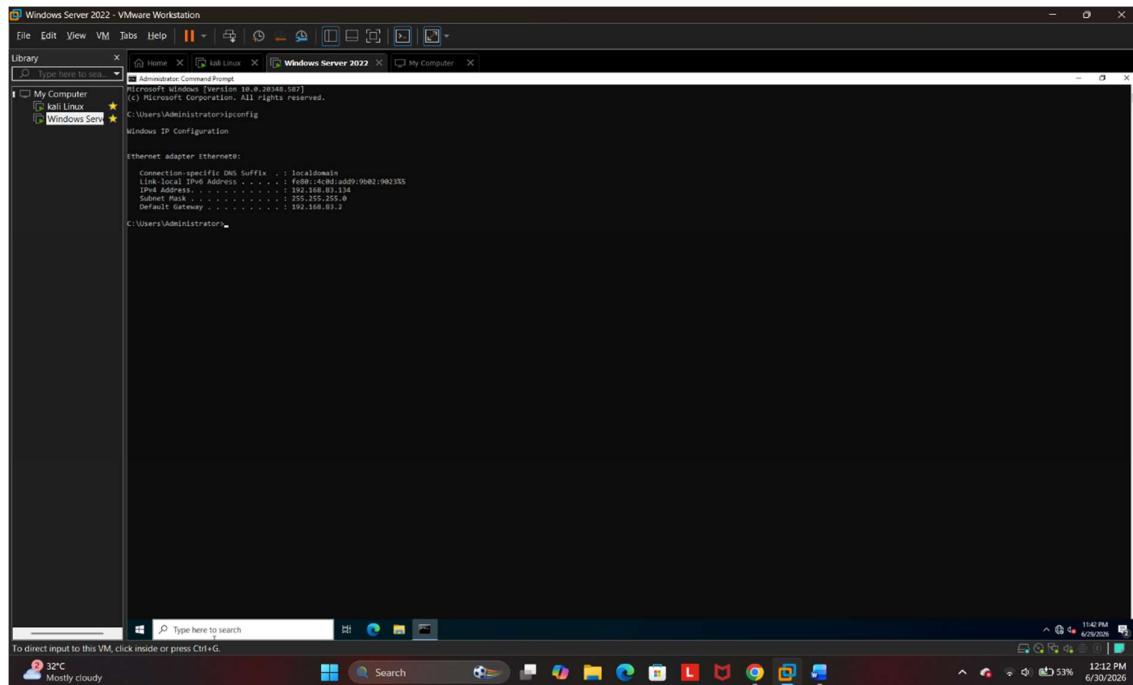


Description:

The "winver" command confirms that Windows Server 2022 is installed.

4. Network Configuration

Figure 4: Windows Server IP Configuration

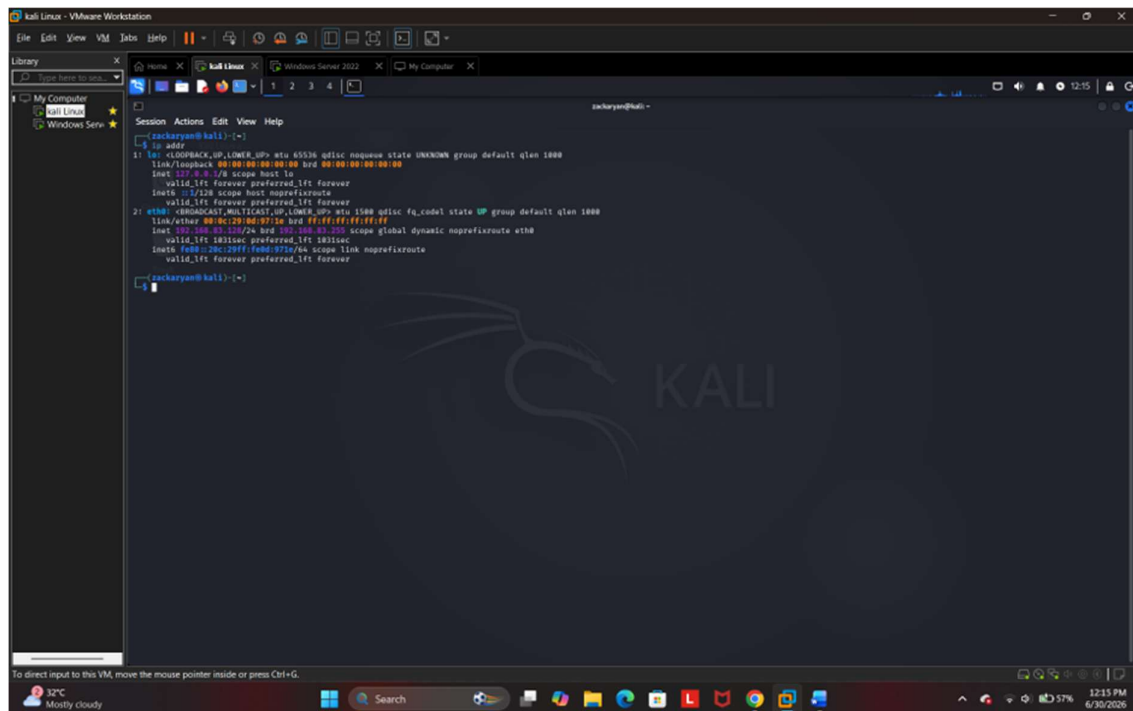


Description:

IP configuration displayed using the ipconfig command.

cmd - ipconfig

Figure 5: Kali Linux IP Configuration



Description:

IP configuration displayed using the ip a command.

Cmd – ip a

Cmd – ip addr

5. Connectivity Testing

Figure 6: Ping Test from Kali Linux to Windows Server

Cmd – ping 192.168.83.134

Cmd – TO stop ping command press ctrl +c

Kali Linux terminal

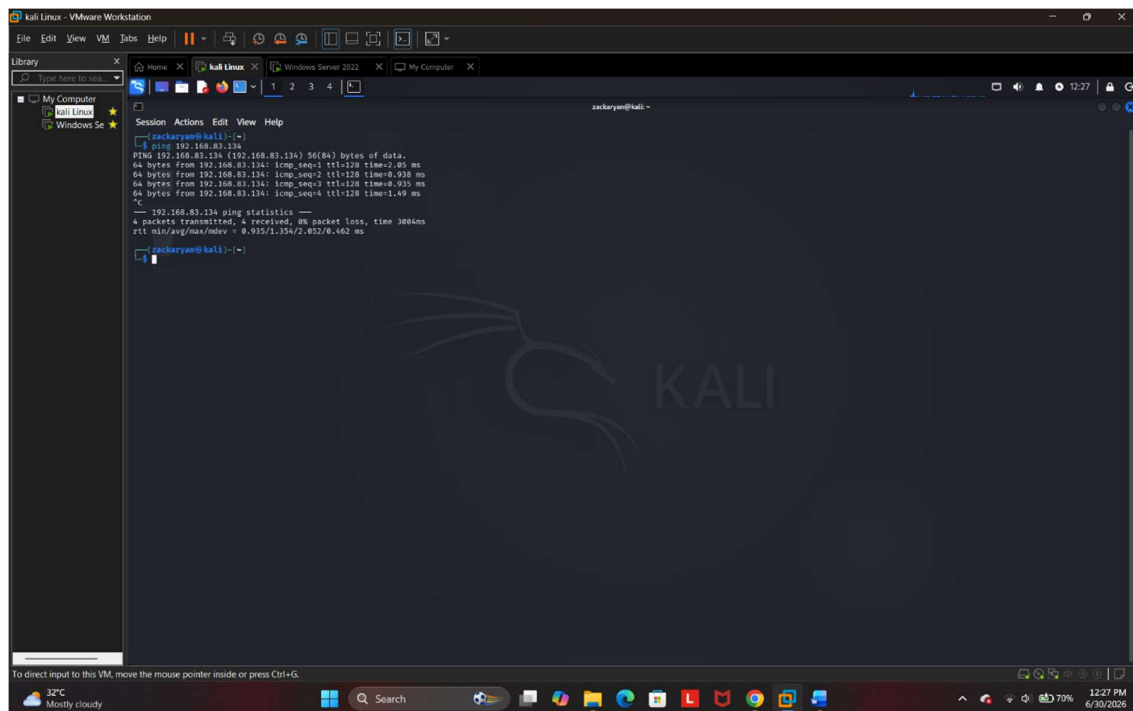
ping 192.168.83.134 command

Multiple successful replies

0% packet loss

Ping statistics

VMware environment

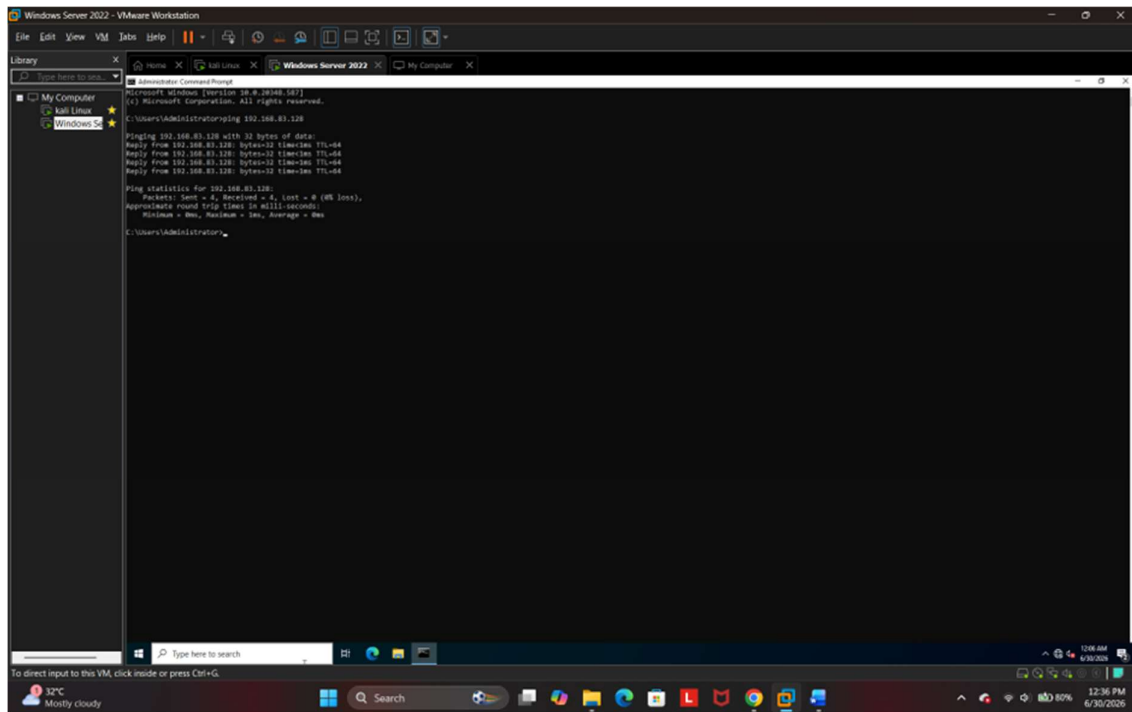


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Description:

ICMP packets successfully transmitted from Kali Linux to Windows Server.

Figure 7: Ping Test from Windows Server to Kali Linux



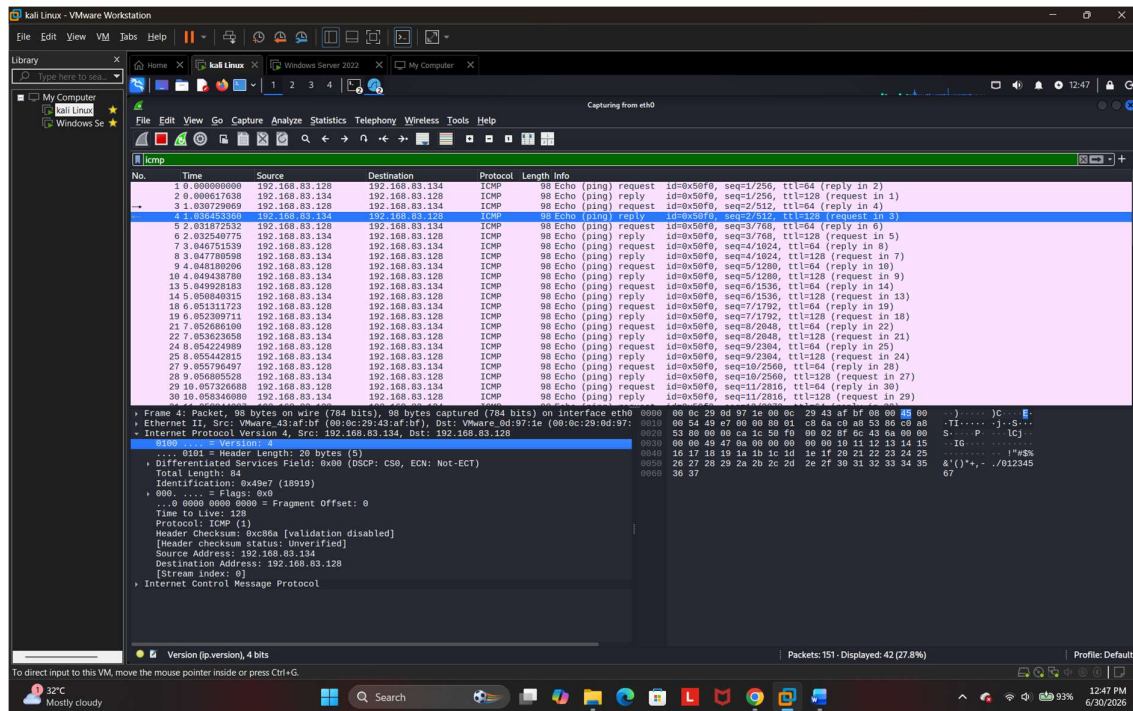
Description:

Successful connectivity confirmed between both virtual machines.

Cmd – ping 192.168.83.128

6. ICMP Traffic Analysis

Figure 8: ICMP Packet Capture



Filter Used:

ICMP

Analysis:

Field Observation

Protocol - ICMP

Source IP - 192.168.83.128

Destination IP - 192.168.83.134

Packet Type - Echo Request / Echo Reply

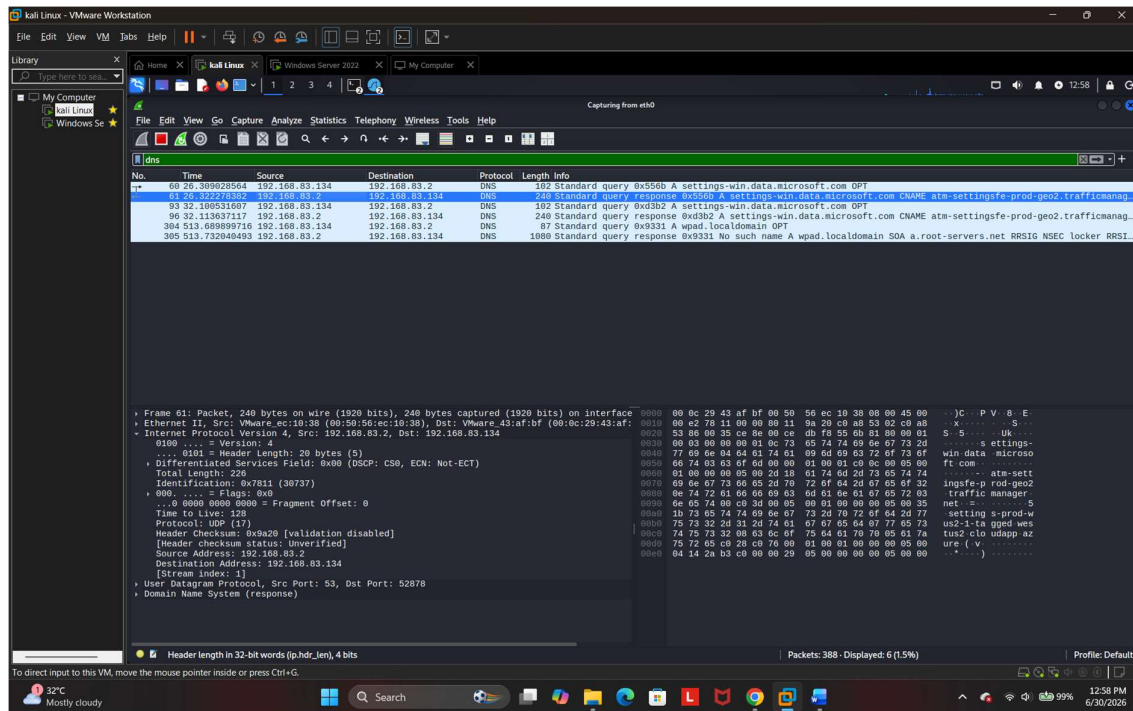
TTL - 64 / 128

Findings:

ICMP packets were captured successfully during ping operations. Echo Request and Echo Reply messages confirmed network connectivity between the virtual machines.

7. DNS Traffic Analysis

Figure 9: DNS Packet Capture



Filter Used:

DNS

Analysis:

Field Observation

Protocol - DNS

Transport Protocol - UDP

Source Port - 53

Query Type - A Record

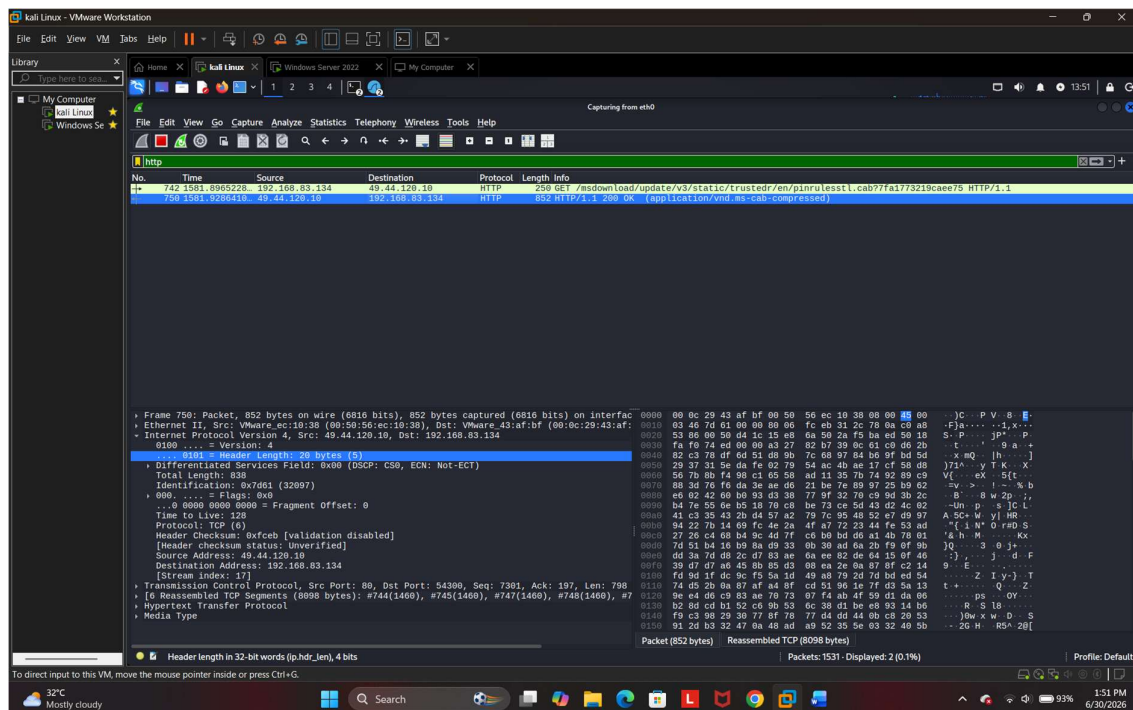
Protocol Purpose Domain Name Resolution

Findings:

DNS packets were captured successfully. The DNS server resolved domain names into IP addresses.

8. HTTP Traffic Analysis

Figure 10: HTTP Packet Capture



Filter Used:

HTTP

Analysis

Field Observation

Protocol - HTTP

Method - GET

Response- HTTP/1.1 200 OK

Transport Protocol- TCP

Source IP - 49.44.120.10

Destination IP – 192.168.83.134

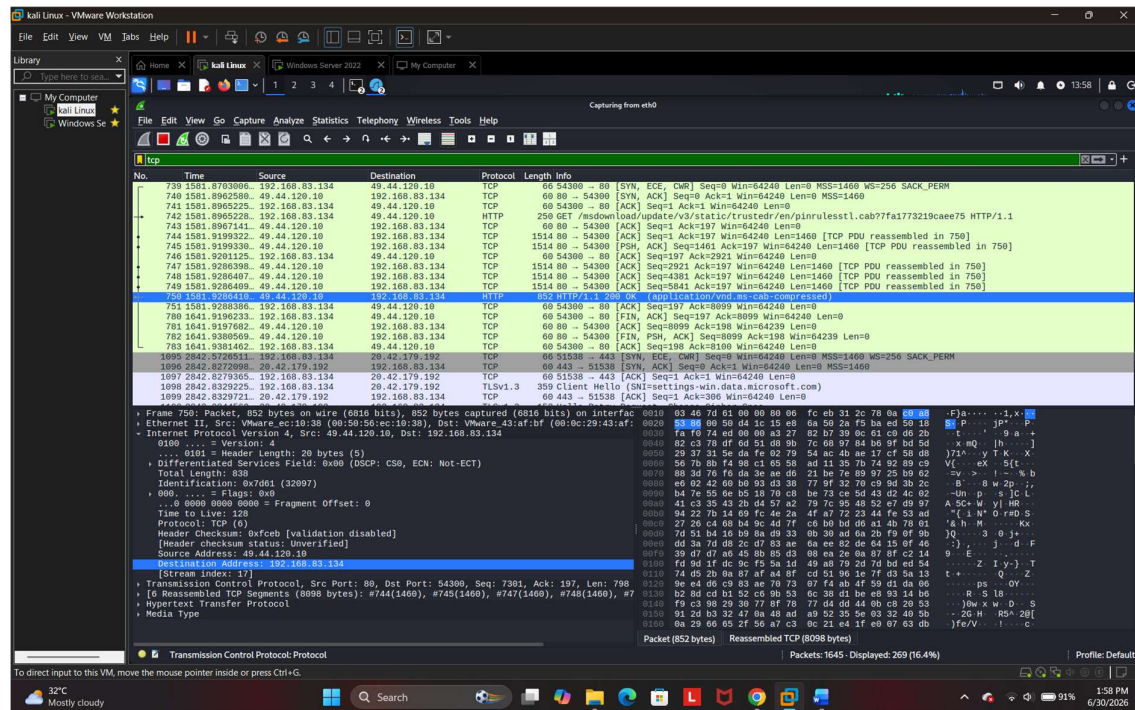
Findings:

HTTP traffic was captured successfully. The packets show communication between a client and a web server using the HTTP protocol.

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9. TCP Protocol Analysis

Figure 11: TCP Packet Capture



Filter Used:

TCP

Analysis:

Field Observation

Protocol - TCP

Source Port - 54300

Destination Port -80

Flags Observed - SYN, SYN-ACK, ACK

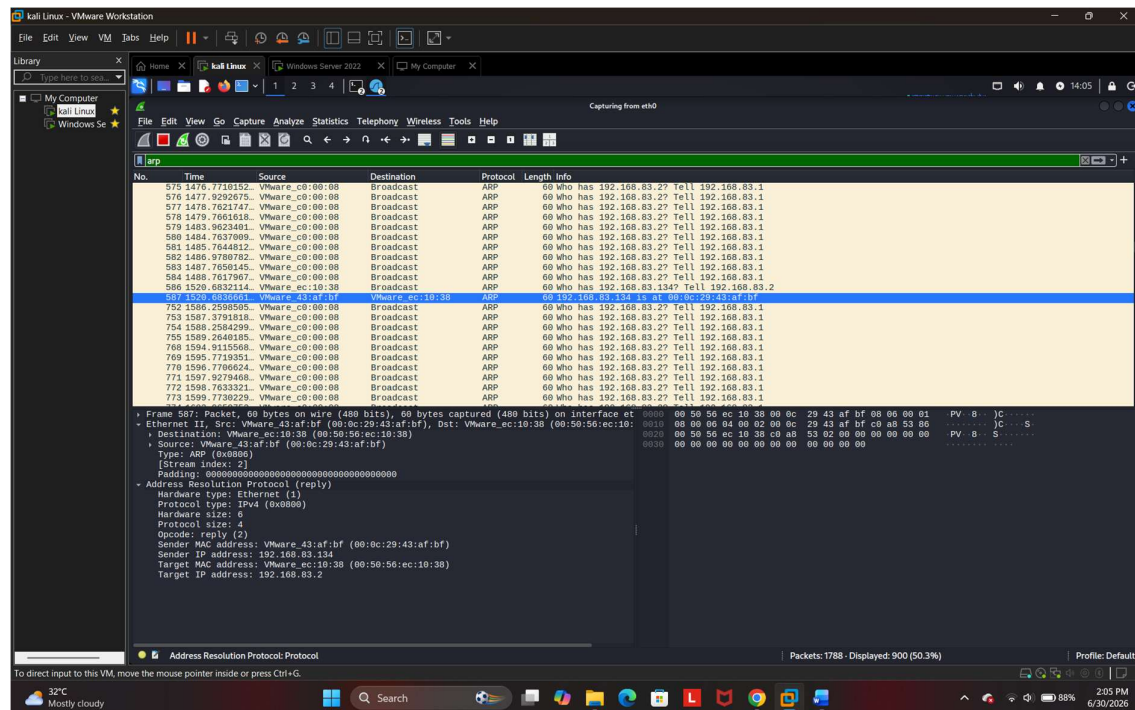
Purpose - Reliable Data Transfer

Sequence Number - 7301

TCP packets demonstrate reliable communication using acknowledgments and sequence numbers.

10. ARP Protocol Analysis

Figure 12: ARP Packet Capture



Filter Used:

ARP

Analysis:

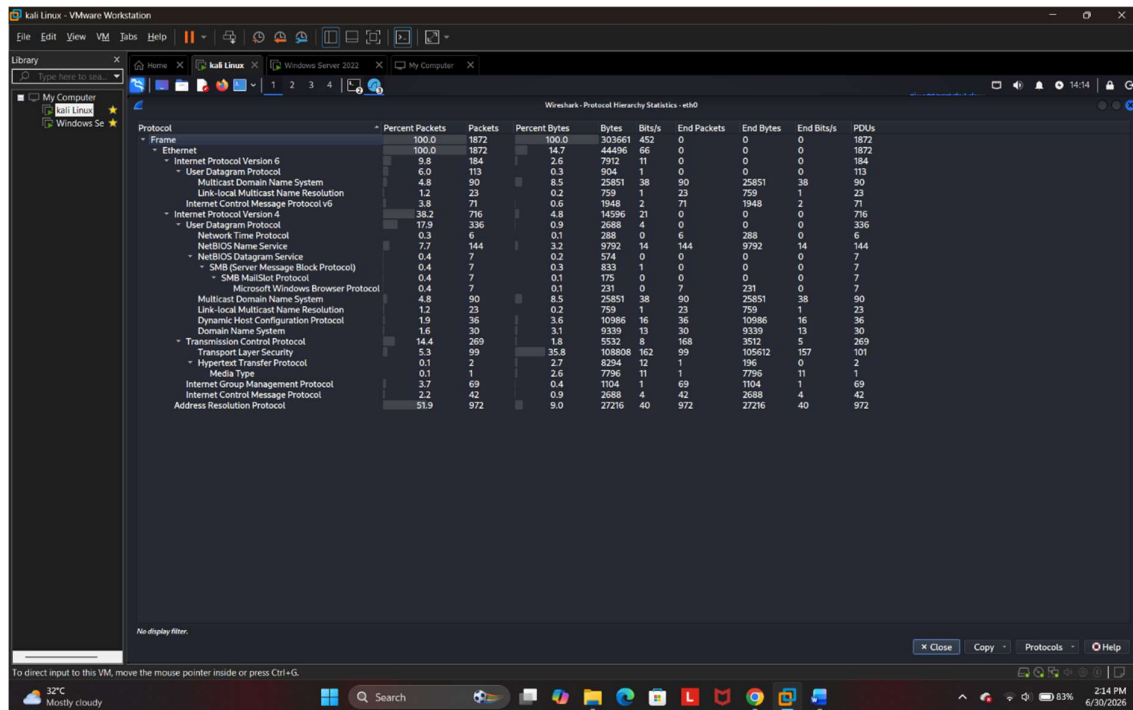
<u>Field</u>	<u>Observation</u>
Protocol -	ARP
Sender IP -	192.168.83.134
Target IP -	192.168.83.2
Operation -	ARP Request / Reply
Purpose -	IP to MAC Address Resolution

Findings: ARP packets were observed resolving IP addresses into MAC addresses for local network communication.

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11. Protocol Hierarchy Statistics

Figure 13: Protocol Hierarchy



Description:

Protocol Hierarchy statistics generated by Wireshark showing the distribution of captured protocols.

12. Protocol Summary

Protocol	Purpose	Observed
ARP	IP-to-MAC address resolution	Yes
ICMP	Connectivity testing	Ping
DNS	Domain name resolution	Yes
TCP	Reliable transport protocol	Yes
HTTP	Web communication	Yes

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13. Results

The home lab network monitoring project was successfully completed using VMware Workstation, Windows Server 2022, Kali Linux, and Wireshark. Network traffic was captured and analyzed for multiple protocols including ARP, ICMP, DNS, TCP, and HTTP. Packet captures confirmed successful communication between virtual machines and external network services.

14. Conclusion

This project demonstrated the setup of a virtual network monitoring environment and the use of Wireshark for packet analysis. The analysis of ICMP, DNS, HTTP, TCP, and ARP traffic provided practical understanding of network communication and protocol behavior. The project successfully met all objectives related to traffic capture, protocol analysis, and documentation.